Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Entropy is not continuous in standard probability spaces

Files.

- File: `briefs/EntropyTopology/EntropyTopology.lean`
- Library: `EntropyTopology`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, `Mathlib.Analysis.SpecialFunctions.Log.Base`, `Mathlib.MeasureTheory.Integral.DominatedConvergence`, `ForMathlib.Analysis.SpecialFunctions.Log.NegMulLog`, `ForMathlib.MeasureTheory.Measure.TotalVariation.Basic`

Setting. A measure space (X, μ) (`MeasureTheory.Measure`). The predicate `IsDensity` says that $\rho\colon X \to \mathbb{R}$ is measurable, pointwise nonnegative, integrable with $\int \rho\, d\mu = 1$, and that $\rho \log \rho$ is integrable. This is the space P_μ; as a type it is `Density`. The entropy S and the Jensen–Shannon divergence are in bits, and $d_1(\rho, \sigma) = \int |\rho - \sigma|\, d\mu$. The probability measure p_ρ with density ρ is `prob` $\mu\, \rho$, and the total variation distance is Mathlib's `tvDist`, i.e. $|p_\rho - p_\sigma|(X)$.

For a divergence D there are two topologies "generated by the balls" $\{\sigma : D(\sigma, \rho) < \delta\}$. In `divTopology` D a set is open iff it contains a ball around each of its points; `ballTopology` D is generated by declaring all balls open. For a metric they agree; in general the second is finer. With D = JSD they are `jsdTopology` and `jsdBallTopology`; `tvTopology` is `divTopology` of $\mathrm{tvDist}(p_\sigma, p_\rho)$. The $L^1(\mu)$ topology `l1Topology` is induced by $\rho \mapsto [\rho] \in L^1(\mu)$ (`Density.toL1`).

Results.

Brief	Lean
$\int_X \lvert\rho - \sigma\rvert \, d\mu$ is the total variation of the measures with densities ρ, σ	`tvDist_withDensity_eq_integral` (ForMathlib)
eq. (2), norm reading: $\lvert p_\rho - p_\sigma\rvert(X) = d_1(\rho, \sigma)$	`tvDist_prob`
eq. (2), event reading: $\sup_A \lvert p_\rho(A) - p_\sigma(A)\rvert = \frac{1}{2} d_1(\rho, \sigma)$, attained at $A = \{\sigma < \rho\}$	`isGreatest_abs_prob_sub`
after eq. (2): L^1 convergence is total-variation convergence	`tendsto_tvDist_iff`
Lemma 4: $u^2/(2 \ln 2) \le f(u) \le \lvert u\rvert$ on $[-1, 1]$	`kernel_bounds`; via $f(u) = 1 - H_2(\frac{1+u}{2})$ (`kernel_eq`)
eqs. (15)–(17), pointwise	`jsdDensity_eq`, `jsdDensity_bounds`
$(\rho + \sigma)/2 \in P_\mu$ (used implicitly)	`isDensity_mid`
$\lambda\rho + (1 - \lambda)\sigma \in P_\mu$ for $\lambda \in [0, 1]$ (presupposed by Theorem 46)	`isDensity_mixture`; pointwise bounds `negMulLog_mixture_bounds`, via $\eta(x + y) \le \eta(x) + \eta(y)$ (`Real.negMulLog_add_le`, ForMathlib)
eq. (16): JSD as one integral	`jsd_eq_integral`
eqs. (18)–(19)	`l1dist_sq_le`: $d_1^2 \le 2 \int (\rho - \sigma)^2/(\rho + \sigma) \, d\mu$
Prop. 10, eq. (11)	`jsd_bounds`
Prop. 10, eq. (12)	`tendsto_jsd_iff`, for any filter
remark after Prop. 10: JSD and total-variation convergence agree	`tendsto_jsd_iff_tvDist`
Prop. 10, last sentence: the JSD topology is the L^1 topology	`jsdTopology_eq` (balls around each point), `jsdBallTopology_eq` (generated by the balls)
the total-variation topology is the L^1 topology	`tvTopology_eq`
$\sigma \mapsto \mathrm{JSD}(\sigma, \rho)$ is L^1-continuous (needed for `jsdBallTopology_eq`)	`continuous_jsd_left`; sequentially in `tendsto_jsd_of_tendsto_l1dist`
eq. (26): $S(u_U) = \log_2 \mu(U)$	`entropy_unif`
eqs. (32)–(34): entropy of $(1 - \varepsilon)u_B + \varepsilon u_A$	`entropy_mix`
eq. (31): $d_1(\rho_n, \rho) = 2\varepsilon_n$	`l1dist_mix`
Theorem 21	`entropy_not_l1_continuous`: $\rho_n \to u_B$ in L^1 and $S(\rho_n) \to S(u_B) + 1$; `not_tendsto_entropy`
second remark, eqs. (40)–(45)	`tendsto_entropy_mix_atTop`: $S(\rho_n) \to \infty$

Brief	Lean
Theorem 46	`eventually_jsd_lt` (JSD balls are τ-neighbourhoods), `eventually_l1dist_lt` (τ is finer than the L^1 topology), `eventually_tvDist_lt` (and than the total-variation topology)
Theorem 46, with the brief's hypothesis	`eventually_jsd_lt_of_mixture`, `eventually_l1dist_lt_of_mixture`, `eventually_tvDist_lt_of_mixture`, via `continuous_mid_of_continuous_mixture`
Theorem 46 for topologies (second proof)	`le_jsdBallTopology`, `le_l1Topology`, `le_tvTopology`: τ is finer; `_of_mixture` forms with the brief's hypothesis

Differences from the brief.

- Densities are functions, not L^1 classes, and are required to be measurable and nonnegative everywhere rather than almost everywhere. "$S(\rho) \in \mathbb{R}$" is read as integrability of $\rho \log \rho$.
- Lemma 4 is proved through the binary entropy. With $p = (1+u)/2$ it says $(2p-1)^2/2 \le \ln 2 - H(p) \le \ln 2\,|2p-1|$. The lower bound comes from concavity of $H(p) + 2(p-\frac{1}{2})^2$, using $H'' = -1/(p(1-p)) \le -4$, which is the brief's second-derivative argument. The upper bound comes from concavity of H, the brief's chord argument.
- Eq. (18) uses Cauchy–Schwarz. The formal proof uses the pointwise inequality $|a-b| \le \frac{(a-b)^2}{(a+b)\,2c} + \frac{c(a+b)}{2}$ with $c = d_1/2$, which gives the same bound $d_1^2 \le 2\int(\rho-\sigma)^2/(\rho+\sigma)$.
- Eq. (12) is formalized through convergence along filters. The topological statements are formalized twice. First through neighbourhoods (`eventually_*`), then as equalities and inequalities of topologies on `Density`. Since JSD is not a metric, "the topology generated by the open balls" is formalized in both readings (see the setting). Theorem 46 then reads $\tau \le$ `jsdBallTopology`, the stronger of the two, which the brief's argument gives directly: continuity of $\sigma \mapsto \mathrm{JSD}(\sigma, \rho)$ makes each ball τ-open. Densities are functions, so these topologies do not separate densities that agree almost everywhere; they are not T_0.
- Eq. (2) is formalized in both readings of $\|\cdot\|_{\mathrm{TV}}$ (see the problem below). The norm reading is proved for any two integrable densities by a Hahn decomposition $\{\rho \le \sigma\}$; no σ-finiteness of μ is needed, which matters since Theorem 21 uses infinite μ.
- In Theorem 21 the hypothesis $1 < \mu(A_n)$ is dropped. Only $\mu(A_n) \to \infty$ is used, and all statements hold eventually in n. The first remark (counting measure) is not formalized separately; it is the special case X countable.
- Theorem 46 is proved in two forms. The main form assumes only that the *equal* mixture $(\rho, \sigma) \mapsto (\rho+\sigma)/2$ is continuous. That is weaker than the brief's hypothesis that all mixing is continuous, so the formal statement is stronger. The `_of_mixture` forms take the brief's hypothesis literally, reading "statistical mixing (e.g. $(\lambda, \rho, \sigma) \mapsto \lambda\rho + (1-\lambda)\sigma$)" as this map $[0,1] \times P_\mu \times P_\mu \to P_\mu$ (`Density.mixture`), with the product topology. They follow by fixing $\lambda = \frac{1}{2}$.

Problems found in the brief. No mathematical errors. One notational ambiguity:

- Eq. (2), $\|p_\rho - p_\sigma\|_{\mathrm{TV}} = \frac{1}{2}\|\rho - \sigma\|_{L^1(\mu)}$, does not say which total variation is meant. With the usual total variation norm $|p_\rho - p_\sigma|(X)$, which is Mathlib's `tvDist`, the factor $\frac{1}{2}$ is wrong: $|p_\rho - p_\sigma|(X) = \|\rho - \sigma\|_{L^1(\mu)}$ (`tvDist_prob`). The factor is right for the distance $\sup_A |p_\rho(A) - p_\sigma(A)|$ (`isGreatest_abs_prob_sub`). Nothing downstream depends on the constant, since only convergence is used.

Gaps the formalization had to fill:

- Prop. 10 and Theorem 46 use $(\rho + \sigma)/2 \in P_\mu$, i.e. that the mixture has finite entropy. The brief does not prove this. It follows from the pointwise bounds $0 \le$ JSD-integrand $\le \frac{\ln 2}{2}|\rho - \sigma|$ (`isDensity_mid`).

- Theorem 46 presupposes that P_μ is closed under mixing, which the brief does not prove either. For $m = \lambda a + (1-\lambda)b$ and $\eta(x) = -x \ln x$, concavity and subadditivity of η give $0 \le \eta(m) - \lambda\eta(a) - (1-\lambda)\eta(b) \le \eta(\lambda)\, a + \eta(1-\lambda)\, b$, so $\eta(m)$ is integrable (`isDensity_mixture`).

- The last sentence of Prop. 10 needs more than eq. (11) in the second reading. The bounds make each JSD ball an L^1 neighbourhood of its centre, which gives `jsdTopology_eq`. For the topology generated by the balls, each ball must be L^1-open, i.e. $\sigma \mapsto \mathrm{JSD}(\sigma, \rho)$ must be L^1-upper semicontinuous; the brief does not address this. The formal proof shows full continuity. For $\sigma_n \to \sigma$ in L^1, a subsequence converges a.e. (via convergence in measure). The integrand g_n is split into $\min(g_n, G)$, with $G = \frac{\ln 2}{2}(\sigma + \rho)$, which converges by dominated convergence, and the excess $g_n - \min(g_n, G)$, which is at most $\frac{\ln 2}{2}|\sigma_n - \sigma|$ by eq. (17) (`tendsto_integral_jsdDensity`).

- In Theorem 21, $\varepsilon_n = 1/\log_2 \mu(A_n) \in (0, 1)$ needs $\mu(A_n) > 2$, not $\mu(A_n) > 1$. The brief's "passing to a tail" covers this.